

Raw Mass Is Not Routing Phase

A normalization correction for antidiagonal optimal-face tomography

30 August 2026

Abstract

A recent optimal-face audit of the quartic zero-repair linear program found a strictly positive joint antidiagonal spread,

$$\text{jointfloor}^* = 1.475854331283591 \times 10^{-2},$$

together with an essentially exact dual certificate. The certificate is mathematically correct, but it reveals that the tested observable was the *raw antidiagonal routing mass*

$$W_h(s, x) = \sum_{j+y=s} z_h(j, y, x),$$

rather than the normalized conditional routing law

$$G_h(x \mid s) = \frac{W_h(s, x)}{R_h(s)}, \quad R_h(s) = \sum_x W_h(s, x).$$

In the reported instance the entire positive joint floor is explained by a difference in total antidiagonal mass:

$$\text{jointfloor}^* = \frac{1}{3} |R_{(3,0)}(2) - R_{(2,1)}(2)|$$

up to approximately 3.5×10^{-15} . Thus the computation rigorously detects unavoidable history dependence of *mass*, but it does not by itself prove unavoidable history dependence of the *normalized routing proportions*. This note isolates the distinction, proves the general pigeonhole certificate behind the observed floor, retracts the stronger phase interpretation, and formulates the correct next tests.

Epistemic convention. Statements of finite-dimensional linear algebra are labelled **PROVED**. Reported solver values are labelled **NUMERICAL**. No finite numerical observation is promoted to an infinite-horizon theorem.

1 Setup

At an internal history h , let

$$z_h(j, y, x) = \Pr(J = j, Y = y, X = x)$$

denote the zero-repair coupling variables. Introduce the antidiagonal coordinate

$$s = j + y.$$

For a fixed s and output symbol x , define the raw routed mass

$$\boxed{W_h(s, x) := \sum_{j+y=s} z_h(j, y, x).} \tag{1}$$

Summing over admissible x gives the total mass incident on the antidiagonal:

$$R_h(s) := \sum_x W_h(s, x) = \sum_{j+y=s} A_h(j)Q(y). \quad (2)$$

Whenever $R_h(s) > 0$, the normalized conditional routing law is

$$G_h(x \mid s) := \frac{W_h(s, x)}{R_h(s)}. \quad (3)$$

Equations (1)–(3) describe two genuinely different notions of history dependence:

$$\begin{array}{ll} \text{mass dependence:} & R_h(s) \text{ depends on } h, \\ \text{routing-phase dependence:} & G_h(\cdot \mid s) \text{ depends on } h. \end{array} \quad (4)$$

The first may occur even when all histories use exactly the same normalized routing proportions.

2 What the joint-floor LP actually certifies

Let $\mathcal{F}^*$ denote the chosen optimal or near-optimal face. For a raw cell $q = (s, x)$, define

$$\text{spread}_q(z) = \max_h W_h(s, x) - \min_h W_h(s, x).$$

The joint-face program computes

$$\text{jointfloor}^* = \min_{z \in \mathcal{F}^*} \max_q \text{spread}_q(z). \quad (5)$$

This is a linear program after introducing upper and lower envelope variables for each q .

The important point is that (5) is a statement about W , not G . A positive value therefore proves that no point of $\mathcal{F}^*$ can make all *raw* routing masses history-independent. It does not yet identify why.

The extracted dual certificate resolves that question in the tested instance.

Proposition 1 (Total-mass pigeonhole bound). *Fix an antidiagonal s with m admissible output symbols $x \in X_s$, and two histories h_+ and h_- . Then*

$$\max_{x \in X_s} |W_{h_+}(s, x) - W_{h_-}(s, x)| \geq \frac{|R_{h_+}(s) - R_{h_-}(s)|}{m}. \quad (6)$$

Consequently,

$$\max_{x \in X_s} \text{spread}_{s,x}(z) \geq \frac{|R_{h_+}(s) - R_{h_-}(s)|}{m}. \quad (7)$$

Proof. Write

$$d_x = W_{h_+}(s, x) - W_{h_-}(s, x).$$

By definition,

$$\sum_{x \in X_s} d_x = R_{h_+}(s) - R_{h_-}(s).$$

Hence

$$|R_{h_+}(s) - R_{h_-}(s)| = \left| \sum_x d_x \right| \leq \sum_x |d_x| \leq m \max_x |d_x|,$$

which proves (6). Since

$$\text{spread}_{s,x}(z) \geq |W_{h_+}(s, x) - W_{h_-}(s, x)|,$$

equation (7) follows. $\square$

Status. Proposition 1 is **PROVED**.

3 The observed certificate

In the depth-3, $M = 8$ audit, the active dual support selected the three cells

$$(s, x) \in \{(2, 0), (2, 1), (2, 2)\}$$

with equal weights

$$\alpha_0 = \alpha_1 = \alpha_2 = \frac{1}{3}, \quad (8)$$

and the history pair

$$h_+ = (3, 0), \quad h_- = (2, 1). \quad (9)$$

Since $x = 0, 1, 2$ are exactly the admissible output symbols on $s = 2$, their sum is the full antidiagonal mass:

$$\sum_{x=0}^2 W_h(2, x) = R_h(2).$$

The extracted certificate therefore reduces to

$$\text{jointfloor}^* \geq \frac{1}{3} |R_{(3,0)}(2) - R_{(2,1)}(2)|. \quad (10)$$

The reported numerical values are

$$\text{jointfloor}^* = 0.01475854331283591, \quad (11)$$

$$\frac{1}{3} |R_{(3,0)}(2) - R_{(2,1)}(2)| = 0.01475854331283239. \quad (12)$$

Their difference is approximately

$$3.5 \times 10^{-15}. \quad (13)$$

The KKT stationarity residual of the extracted dual certificate is of the same order.

Status. Equations (11)–(13) are **NUMERICAL**. They show, to essentially machine precision, that the measured joint floor is saturated by the total-mass pigeonhole obstruction.

This is an important *explanation* of the finite LP value, but it changes the interpretation of that value.

4 Correction of the phase interpretation

The previously tempting interpretation was:

A positive joint floor proves that some history-dependent fresh routing phase survives quotienting by optimal-face degeneracy.

That statement is too strong for the observable actually tested.

Indeed, suppose for some fixed probability vector $p_s(x)$ that

$$W_h(s, x) = R_h(s) p_s(x) \quad \text{for every history } h. \quad (14)$$

Then

$$G_h(x \mid s) = p_s(x)$$

is completely history-independent, while the raw masses $W_h(s, x)$ still vary whenever $R_h(s)$ varies.

Thus

$$R_{h_+}(s) \neq R_{h_-}(s)$$

can force a strictly positive raw joint floor even in a world with *zero normalized routing phase*.

The corrected conclusion is therefore

$$\boxed{\text{jointfloor}^*(W) > 0 \implies \text{unavoidable history dependence of raw antidiagonal mass,}} \quad (15)$$

but not

$$\boxed{\text{jointfloor}^*(W) > 0 \not\Rightarrow \text{unavoidable history dependence of } G_h(\cdot | s).} \quad (16)$$

Status. The implication distinction (15)–(16) is **PROVED**.

5 Why this does not make the computation useless

The joint-floor computation remains mathematically valuable for three reasons.

First, it correctly exposed a nontrivial fact about the entire optimal face: the raw antidiagonal mass profile cannot be made simultaneously history-independent.

Second, the dual extractor did more than reproduce a number. It identified the exact finite-dimensional mechanism behind the floor: a three-channel partition of one antidiagonal together with a total-mass imbalance between two histories.

Third, the correction cleanly separates two levels of structure that should not be conflated:

$$\boxed{A_h \longrightarrow R_h(s) \longrightarrow W_h(s, x) = R_h(s)G_h(x | s).} \quad (17)$$

History dependence can enter already at the A_h or R_h level. A claim of genuinely new routing phase must survive after that amplitude dependence has been controlled.

6 The correct next tests

6.1 A linear mass-controlled slice

For a selected pair of histories h_+, h_- and antidiagonal s , impose the additional linear equality

$$\boxed{R_{h_+}(s) = R_{h_-}(s).} \quad (18)$$

If the resulting face slice is nonempty, then any remaining raw routing difference between those two histories cannot be explained by a difference in their total antidiagonal mass.

Define

$$\Phi_{h_+, h_-; s}^{\text{eqmass}} := \min_{\substack{z \in \mathcal{F}^* \\ R_{h_+}(s) = R_{h_-}(s)}} \max_{x \in X_s} |W_{h_+}(s, x) - W_{h_-}(s, x)|. \quad (19)$$

After the usual epigraph linearization, (19) is an LP.

If

$$\Phi_{h_+, h_-; s}^{\text{eqmass}} > 0,$$

then normalized routing differs between h_+ and h_- on that equal-mass slice, because their denominators are identical.

This is a direct and inexpensive falsification test.

6.2 Direct normalized routing

The intrinsic quantity is

$$G_h(x \mid s) = \frac{W_h(s, x)}{R_h(s)}.$$

Single-ratio extrema such as

$$\min_{z \in \mathcal{F}^*} G_h(x \mid s), \quad \max_{z \in \mathcal{F}^*} G_h(x \mid s)$$

are linear-fractional programs and may be converted to LPs by a Charnes–Cooper transformation when $R_h(s) > 0$ is controlled.

The genuinely history-comparative condition

$$G_h(x \mid s) = G_g(x \mid s)$$

is equivalent to

$$W_h(s, x)R_g(s) = W_g(s, x)R_h(s),$$

which is bilinear. A complete optimal-face test of simultaneous equality of normalized routing therefore requires either a dedicated bilinear/global optimization formulation or a suitable exact reformulation.

Status. The equal-mass LP (19) is a concrete **PROVED** reduction. A complete normalized-routing face tomography is **OPEN**.

7 Revised interpretation of the “ghost” cell

The same normalization warning applies to the previously reported positive floor at

$$(j, y, x) = (6, 1, 4).$$

A positive raw spread there proves that the corresponding coupling mass cannot be made history-independent over the tested face. It does not, by itself, separate a change in total incident mass from a change in normalized routing preference.

Accordingly, the phrase “fresh causal phase is proved to exist” should not be used for that observation unless the effect survives a mass-controlled or normalized-routing test.

The safe statement is:

The finite optimal face contains solver-independent history dependence in raw coupling mass at the reported cell. Whether this represents genuinely history-dependent normalized routing remains unresolved.

8 Implication for the upper-bound program

This correction does not settle the logarithmic upper bound in either direction.

A history-dependent $R_h(s)$ may be entirely compatible with a bounded or logarithmic total bill. Conversely, even a small normalized-routing phase could in principle carry a large transport cost. The upper-bound question therefore cannot be decided from raw joint floors alone.

The useful consequence is methodological:

Any future antidiagonal “phase” observable must quotient or control the total-mass coordinate $R_h(s)$.
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(20)

Only the residual structure after this normalization is a candidate for a genuinely new causal routing obstruction.

9 Research protocol

For future finite-depth audits we propose the following order.

1. Compute the raw optimal-face observable $W_h(s, x)$.
2. Compute and report $R_h(s) = \sum_x W_h(s, x)$ on the same face.
3. Before interpreting a positive raw spread as routing phase, test the equal-mass slice (18).
4. Where feasible, compute direct linear-fractional bounds on $G_h(x \mid s)$.
5. Promote a “fresh phase” claim only if history dependence survives normalization by $R_h(s)$.
6. Keep the raw-mass and normalized-routing ledgers separate.

This protocol prevents amplitude variation from being misidentified as phase variation.

10 Bottom line

The joint-floor dual certificate is not wrong. It is, in fact, sufficiently clean to diagnose the earlier overinterpretation.

What was proved numerically on the finite face is

unavoidable history dependence of raw antidiagonal mass.

What remains to be proved is

unavoidable history dependence of normalized antidiagonal routing.

The distinction is exactly

$$W_h(s, x) = R_h(s) G_h(x \mid s).$$

The next experiment should therefore not ask whether W differs across histories. It should ask whether any difference survives after the R -coordinate has been removed.